EasyPointer: What You Pointing at is What You Get

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Abstract
This extended abstract presents a natural pointing system using gyroscope MEMS, which could achieve *what you pointing at is what you get*. It enables screen interaction via *physical pointing*. The system only needs a phone with built-in gyroscope, camera, and wireless communication such as Wi-Fi and Bluetooth, without any other hardware. To achieve sensing of the screen position where user is physically pointing, we also develop easy calibration methods. The prototype system, called *EasyPointer*, is built upon iPhone 4/iPod 4. It can serve as a laser pointer, a presentation controller, a game controller, and a drawing pen.

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Introduction
Mobile phone is serving as pervasive handheld device in our daily life [6]. In addition to its increasing computing power, sensors on phones are evolving faster than ever. Now many built-in sensors are becoming the standard configuration, such as GPS, accelerometer, camera, and gyroscope. It has been attracted researchers much attention that employ of these built-in sensors to achieve natural interaction.

Pointing interaction has drawn considerable research interests from both academia and industry [4, 8].For example, Wii Remote [9] has built-in infrared filtered camera to measure orientation of itself, but it has to be used with a sensor bar. And finger pointing can be also captured by Microsoft Kinect [10], which employed infrared light with structural pattern. But Kinect suffers the same problem of requirement of additional accessory. AirMouse is a portable wireless mouse enabling in air pointing, but it is inconsistent with what users actually

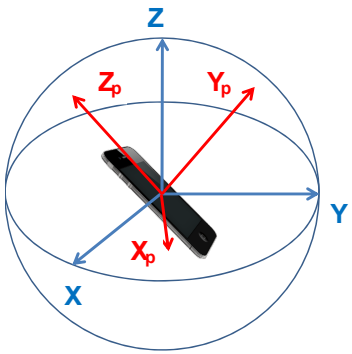

figure 1. Illustration of the pointing orientation of a phone in the world coordinate system

figure 2. Taking a shot of screen for calibration. The left-bottom corner shows the detected screen region in the image.

point to the screen. Recent literature [3] investigated how constrained hand moment can be used for pointing. Other work such as [2] has already explored the concept of using phones as pointing devices using camera. But this approach required the camera always on. Our previous work, MagicPhone [5], is also a pointing device but it focus on appliance selection.

This extended abstract presents our ongoing work on physical pointing interaction on a screen using a mobile phone with built-in gyroscope. It can achieve *what you pointing at is what you get* without any other accessories.

Interaction via Phone-based Natural Pointing

Based on the principle of conservation of angular momentum, a mechanical gyroscope is basically a wheel spinning in high rate with its axle orientation free. A given external torque has little effect on it due to its large angular momentum, so that its orientation can be maintained, no matter how mounted platform moves or rotates. To present, the built-in gyroscopes of mobile phones, such as iPhone 4 and Nexus S, are all manufactured with Micro-electro-mechanical systems (MEMS) technology. In essence, MEMS gyroscope is integrated circuits packaged vibrating structures for multiple (e.g. 3) axes. These vibrating structures are insensitive to how much the mounted platform rotated. By measuring the change of vibration inside in related direction, rotation velocity of mounted platform can be calculated.

Gyroscopic MEMS provides a mobile phone with self-awareness of its pose. Three Euler angles, *pitch*, *yaw*, and *roll* (α, β, γ), can be extracted from gyroscope data.

Initially the pointing orientation of a mobile phone (the positive direction of z_p-axis of the phone coordinate system), illustrated in figure 1, can be represented by rotating the *Z*-axis in the world coordinate system xyz with an Euler angle. The *roll* component will count nothing about the pointing orientation. Our pointing problem is reduced to mapping pitch and yaw values of a device orientation to 2-D screen.

Calibration of Pointing Orientation

The main difficulty of natural pointing interaction is how we can know where user is pointing to. For the pointing interaction with a screen, that is how to automatically map the physical pointing orientation to the corresponding position on the screen. The key problem is to tackle variation of screen size and variation of user position relative to a screen, since both user moving and different screen size will affect the mapping. To cope with these, a calibration process of pointing orientation is required to enable what you pointing at is what you get.

The physical pointing orientation is described by the *pitch* and *yaw* of a phone: (α, β), which can be measured by gyroscope MEMS built-in a phone. Suppose that (x, y) is a point on the screen, the calibration problem is converted to find the mapping

$$\mathcal{M}: (\alpha, \beta) \rightarrow (x, y)$$

that satisfies (x, y) physically lies on the orientation (α, β) , i.e. (x, y) is the intersection of the screen plane and the ray with the orientation (α, β)originated from the phone. We develop three calibration methods:

Method 1: Center-based calibration

figure 3. Playing games with EasyPointer.

The center-based calibration is essentially an approximate approach, which based on three simplified assumptions: 1) a user stands right in front of the screen, 2) the device and screen center are on same altitude; and 3) rough size of screen and distance between user and screen are known. Initially user points to the center of screen, and then the screen center is mapping to current device orientation, to accomplish calibration.

Method 2: Four-corner-based calibration
Pointing to four corners of screen in turn can be used to calibrate the effective intervals of orientation, namely, pitch and yaw. Arbitrary device orientation falling in these intervals can be approximately obtained by mean of interpolation according to four corners is one of alternative in addition to precise geometric approach. It is hard for users to accurately point to a.

Method 3: Calibration by taking a picture
For a place where a user is standing, this method only needs user take a shot of the screen to accomplish calibration of pointing orientation. We assume the imaging system of phone camera is a pinhole model, and the image resolution is assumed to be fixed. The calibration process can be carried out by three steps:

1) Take a shot of the screen by the phone camera, and simultaneously save the phone pointing orientation at the shooting time, shown in figure 2.
2) Detect the four corners of the screen in the image by computer vision algorithm [7].
3) Compute the normal of the screen plane [1]

EasyPointer: Implementation & Features

The *EasyPointer* uses iPhone 4/iPod touch 4 as the pointing device, which has built-in 3D gyroscopic sensor and Wi-Fi module. The pointing device is wirelessly connected the host computer of a screen to interact with the screen. The screen could be computer monitor and projector screen. Physical pointing interaction using EasyPointer is very intuitive. It could be extended as a universal controller and pointing device in various applications. EasyPointer features several advantages:

EasyPointer as a Presentation Assistant

When a lecturer is delivering a talk, it can serve as a laser pointer with any "laser" color to highlight something of interest in the slide, via the real pointing like using laser pointer. It can page-down and page-up the slides remotely. It can also weave the EasyPointer in air to draw figures and curves in the slide.

EasyPointer as a Game Controller

It can offer extra benefits with better immersion when gaming, especially in First Person Shooter game. However, while optical mice require a surface to operate, EasyPointer free users from these constraints and they stand or sit on wherever they feel pleased and the gaming experience can be enhanced.

EasyPointer as a Drawing Pen

EasyPointer is also suitable for users to make their artworks with painting in air, since it resembles a physical pen. It supports multi-users to draw cooperatively. Children drawing education is one of potential scenarios.

Conclusions

Toward an ideal pointing device with *what you pointing at is what you get*, we implemented a phone-based

prototype system mainly using a built-in gyroscopic sensor for read and physical pointing interaction. It can be used as a presentation assistant, a game controller, and a drawing pen. Its intuitive pointing manner makes it very promising for various applications.

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